MICHIGAN STATE UNIVERSITY

Michigan State University **Formula Racing Team**

College of Engineering
Electrical and Computer Engineering

Component Load and Direction Sensing System Validation and Testing

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Abstract

This project presents the development and validation of a modular, low-cost load and direction-sensing system designed for Formula SAE vehicle testing. The system provides an adaptable alternative to expensive commercial sensors by integrating strian gauge-based load sensing with uniaxial and triaxial accelerometers to capture dynamic force magnitude and direction.

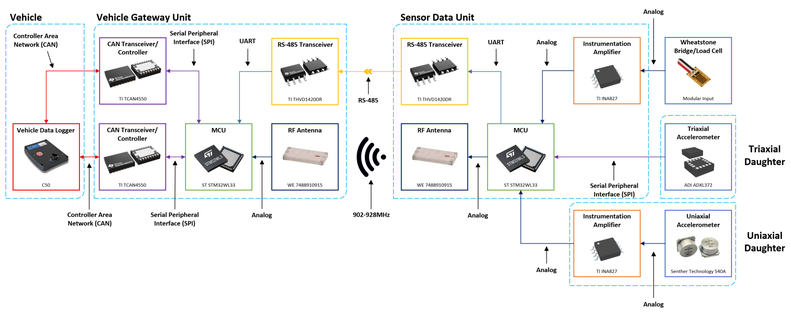

Figure 1: System communication architecture

Testing & Correlation

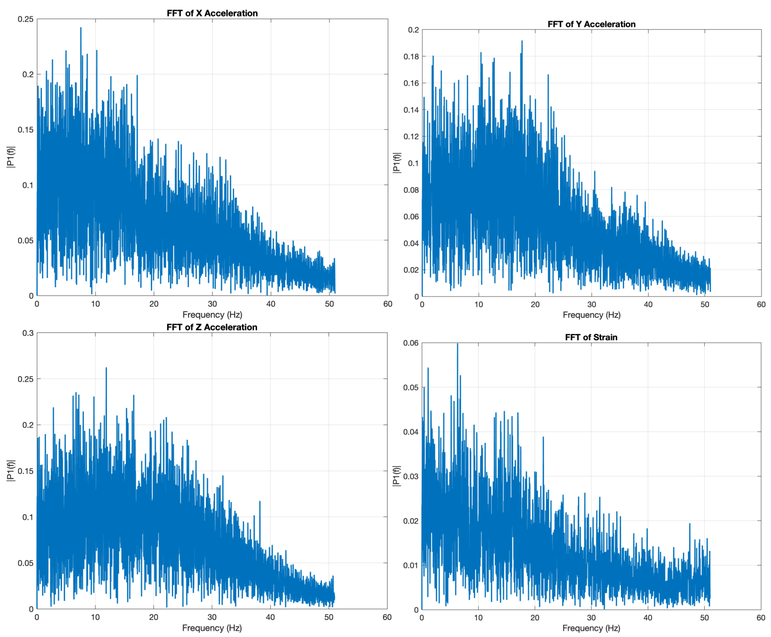

Figure 2: Fast Fourier Transform (FFT) of triaxial acceleration and strain of plate during drop test

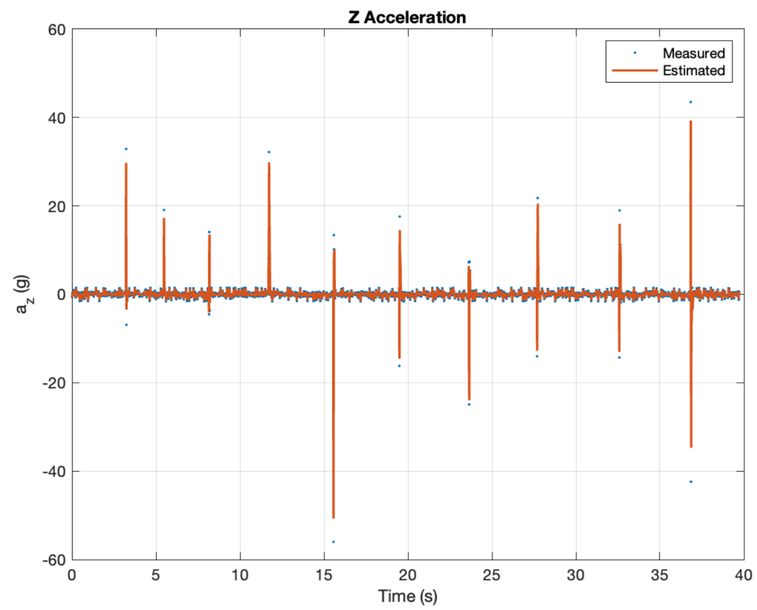

Figure 3: Z-direction acceleration of plate during drop test

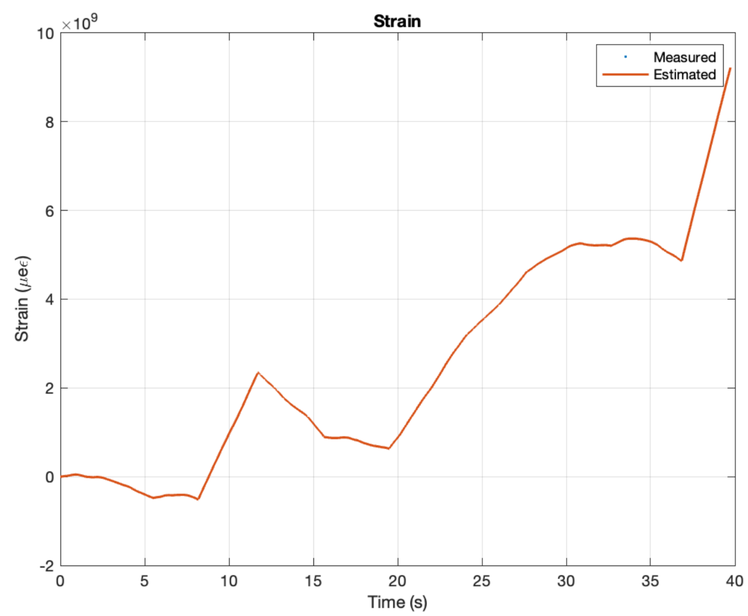

Figure 4: Strain of plate during drop test

This correlation demonstrates that the system successfully integrates strain and inertial measurements to provide a comprehensive understanding of loading conditions.

Performance Evaluation

Specification Category	Requirement	Observed Performance	Status
Strain Measurement	Detect Structural Deformation	Correct trend observed, incorrect magnitude scaling	Partial
Acceleration Measurement	Capture Multi-Axis Motion	X, Y, and Z signals captured successfully	Met
Load Contextualization	Correlate Strain and Motion	Strong correlation observed	Met
Vibration Characterization	Identify Dynamic Response	Clear oscillatory behavior detected	Met
Frequency Analysis	Identify Vibration Modes	Peaks observed (∼15-35Hz)	Met
Data Processing	Filter and Interpret Signals	Kalman filter improves signal quality	Met
Communication	Transmit data reliably	RS-485 and CAN functional	Met

Table 1: Performance evaluation of key requirements

- **Dynamic Response:** Captured synchronized transient strain and 3-axis acceleration. Peak acceleration values reached **41.5g (Y-axis)** and **56.0g (Z-axis)**, with clear oscillatory decay.
- **Signal Processing:** The **Kalman Filter** significantly enhanced data quality, reducing noise standard deviation by **17.0% (Y-axis)** and **18.5% (Z-axis)** while maintaining a high correlation (**>0.95**) with raw physical events.
- **Vibration Characterization:** Frequency-domain analysis (FFT) identified dominant structural vibration modes between **15–35 Hz**, matching the expected natural frequencies of the test apparatus.
- **System Performance:** Achieved high-fidelity data acquisition with sampling rates exceeding **100 kS/s**, ensuring no loss of transient data during impact events.

Cost Breakdown

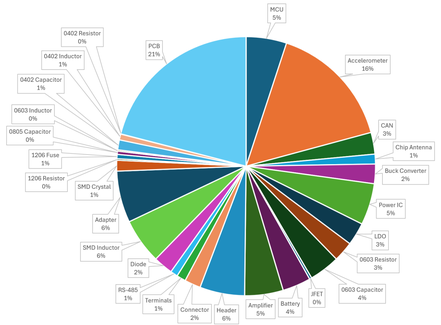

Figure 5: Total project cost breakdown

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